

1. A ratio is used to compare _____.

- a) Similar quantities
- b) Dissimilar quantities
- c) Colours
- d) Shapes

Answer: Option (a) – Similar quantities

2. The symbol used to represent ratio is _____.

- a) $\times$
- b) $\div$
- c) $:$
- d) $::$

Answer: Option (c) – $:$

3. The ratio of 4 to 3 is written as _____.

- a) $4 \div 3$
- b) 4:3
- c) 3:4
- d) $4 + 3$

Answer: Option (b) – 4:3

4. Two ratios are said to be in proportion when _____.

- a) They are equal
- b) They are opposite
- c) They are different
- d) They have different units

Answer: Option (a) – They are equal

5. The symbol used to show proportion is _____.

- a) $\times$
- b) $::$
- c) $=$
- d) Both (b) and (c)

Answer: Option (d) – Both (b) and (c)

6. Ratios can be compared only when the quantities are in the _____.

- a) Same unit
- b) Different unit
- c) Any unit
- d) No unit

Answer: Option (a) – Same unit

7. A ratio has _____ unit.

- a) One
- b) Two
- c) No
- d) Variable

Answer: Option (c) – No

8. The ratio of 10 km to 100 m in simplest form is _____.

- a) 10:1
- b) 100:1
- c) 1:10
- d) 1:100

Answer: Option (b) – 100:1

9. The first quantity in a ratio is called the _____.

- a) Antecedent
- b) Consequent
- c) Multiplier
- d) Denominator

Answer: Option (a) – Antecedent

10. The second quantity in a ratio is called the _____.

- a) Antecedent
- b) Consequent
- c) Product
- d) Dividend

Answer: Option (b) – Consequent

11. To simplify a ratio, we divide both terms by their _____.

- a) Difference
- b) Common factor
- c) Product
- d) Sum

Answer: Option (b) – Common factor

12. To simplify the ratio 45:6, we divide by _____.

- a) 2
- b) 3
- c) 6
- d) 9

Answer: Option (b) – 3

13. The simplified form of 45:6 is _____.

- a) 15:2
- b) 9:1
- c) 7:3
- d) 6:2

Answer: Option (a) – 15:2

14. The first step in simplifying a ratio of same units is to write it in _____ form.

- a) Word
- b) Fraction
- c) Table
- d) Symbol

Answer: Option (b) – Fraction

15. To compare 5 m and 700 cm, we convert 5 m into _____.

- a) mm
- b) km
- c) cm
- d) m²

Answer: Option (c) – cm

16. 5 m = _____ cm.

- a) 50
- b) 500
- c) 5000
- d) 5

Answer: Option (b) – 500

17. The ratio 5 m : 700 cm in simplest form is _____.

- a) 7:5
- b) 5:7
- c) 10:7
- d) 7:10

Answer: Option (b) – 5:7

18. Equivalent ratios can be obtained by _____ both terms by the same number.

- a) Adding
- b) Subtracting
- c) Multiplying or dividing

d) Ignoring

Answer: Option (c) – Multiplying or dividing

19. The ratios 1:2, 2:4, and 3:6 are _____.

a) Equal

b) Different

c) Opposite

d) Equivalent

Answer: Option (d) – Equivalent

20. The ratio 2:4 is obtained by multiplying 1:2 by _____.

a) 1

b) 2

c) 3

d) 4

Answer: Option (b) – 2

21. The ratio 3:6 is obtained by multiplying 1:2 by _____.

a) 2

b) 3

c) 4

d) 6

Answer: Option (b) – 3

22. Equivalent ratios have the same _____ form.

a) Fraction

b) Colour

c) Unit

d) Shape

Answer: Option (a) – Fraction

23. To check if ratios are equivalent, we write them as _____.

a) Decimals

b) Fractions

c) Words

d) Tables

Answer: Option (b) – Fractions

24. Ratios 1:4 and 4:8 are _____.

a) Equal

b) Equivalent

c) Not equivalent

d) Same

Answer: Option (c) – Not equivalent

25. The LCM of 4 and 8 is _____.

a) 4

b) 8

c) 12

d) 16

Answer: Option (d) – 16

26. $\frac{4}{16}$ is _____ than $\frac{8}{16}$.

a) Greater

b) Lesser

c) Equal

d) Same

Answer: Option (b) – Lesser

27. When denominators are equal, the fraction with the greater numerator is _____.

a) Smaller

b) Greater

c) Equal

d) Zero

Answer: Option (b) – Greater

28. $\frac{1}{3}$ is _____ than $\frac{1}{4}$.

a) Greater

b) Smaller

c) Equal

d) None

Answer: Option (a) – Greater

29. Reena completed _____ fraction of her homework than Ravi.

a) Less

b) Equal

c) More

d) None

Answer: Option (c) – More

30. If a rope of 8 m is cut into 3 m and 5 m parts, the ratio of the two parts is _____.

a) 5:3

b) 3:5

c) 8:3

d) 5:8

Answer: Option (b) – 3:5

31. If two ratios are equal, they are said to be in _____.

a) Division

b) Subtraction

c) Proportion

d) Equation

Answer: Option (c) – Proportion

32. The ratios 2:4 and 8:16 are _____.

a) In proportion

b) Not in proportion

c) Different

d) Opposite

Answer: Option (a) – In proportion

33. The ratios 9:4 and 18:6 are _____.

a) In proportion

b) Not in proportion

c) Equal

d) Similar

Answer: Option (b) – Not in proportion

34. The law stating product of extremes = product of means is called _____.

a) Proportionality law

b) Ratio law

c) Multiplication law

d) Fraction law

Answer: Option (a) – Proportionality law

35. In a proportion $a:b::c:d$, 'a' and 'd' are called _____.

a) Means

b) Extremes

c) Factors

d) Values

Answer: Option (b) – Extremes

36. In a proportion $a:b::c:d$, 'b' and 'c' are called _____.

a) Means

b) Extremes

c) Constants

d) Numbers

Answer: Option (a) – Means

37. The cross-product form of proportion is _____.

a) $ab=cd$

b) $ad=bc$

c) $a+b=c+d$

d) $a-b=c-d$

Answer: Option (b) – $ad=bc$

38. For 7:2 and 21:6, $a=7$, $b=2$, $c=21$, $d=6$. The product of extremes =

_____.

a) 42

b) 36

c) 28

d) 12

Answer: Option (a) – 42

39. For 7:2 and 21:6, the product of means = _____.

a) 28

b) 42

c) 36

d) 14

Answer: Option (b) – 42

40. Since $ad = bc$ in 7:2 and 21:6, the ratios are _____.

a) Not proportional

b) In proportion

c) Different

d) Opposite

Answer: Option (b) – In proportion

41. The method used to find value of one unit and then many units is called

_____.

a) Ratio method

b) Unitary method

c) Division method

d) Algebraic method

Answer: Option (b) – Unitary method

42. In the unitary method, first we find the value of _____.

a) Two units

- b) One unit
- c) All units
- d) No unit

Answer: Option (b) – One unit

43. The cost of 4 apples is ₹100. The cost of 1 apple is _____.

- a) ₹10
- b) ₹20
- c) ₹25
- d) ₹40

Answer: Option (c) – ₹25

44. If 1 apple costs ₹25, the cost of 10 apples is _____.

- a) ₹100
- b) ₹200
- c) ₹250
- d) ₹400

Answer: Option (c) – ₹250

45. The first step in the unitary method is to _____.

- a) Convert the problem into mathematical form
- b) Draw a graph
- c) Add the numbers
- d) Write a story

Answer: Option (a) – Convert the problem into mathematical form

46. The second step in the unitary method is to find the value of _____.

- a) One unit
- b) Two units
- c) Three units
- d) All units

Answer: Option (a) – One unit

47. The third step in the unitary method is to find _____.

- a) Required quantity
- b) Mean
- c) Product
- d) Ratio

Answer: Option (a) – Required quantity

48. Ratios are used in _____ life situations.

- a) Scientific only
- b) Day-to-day

- c) Historical
- d) None

Answer: Option (b) – Day-to-day

49. When two numbers are in the ratio $a:b$, they can be written as _____.

- a) a/b
- b) ax and bx
- c) $a+x$ and $b+x$
- d) $a-x$ and $b-x$

Answer: Option (b) – ax and bx

50. For the ratio $8:13$, it can also be written as _____.

- a) $8x:13x$
- b) $8+13$
- c) $8/13x$
- d) $13:8x$

Answer: Option (a) – $8x:13x$

51. The law $ad = bc$ is called

- a) ratio law
- b) equality law
- c) proportionality law
- d) multiplication law

Answer: c) proportionality law

52. The term used for 'comparison of quantities of same kind' is

- a) difference
- b) ratio
- c) average
- d) unit

Answer: b) ratio

53. Ratio has

- a) unit
- b) no unit
- c) two units
- d) three units

Answer: b) no unit

54. 10 km to 100 m ratio after converting to same unit is

- a) $10:100$
- b) $10000:100$
- c) $1:1$

d) 1:10

Answer: b) 10000:100

55. Simplest form of 10000:100 is

a) 10:1

b) 100:1

c) 1:10

d) 1000:1

Answer: b) 100:1

56. To simplify a ratio, we express it in the form of

a) addition

b) subtraction

c) fraction

d) percentage

Answer: c) fraction

57. Simplest form of 45:6 is

a) 15:2

b) 10:2

c) 3:1

d) 9:2

Answer: a) 15:2

58. Simplify 5 m : 700 cm.

a) 5:7

b) 7:5

c) 5:70

d) 70:5

Answer: a) 5:7

59. In simplifying ratios of different units, first step is to

a) divide directly

b) convert to same unit

c) simplify numbers

d) find LCM

Answer: b) convert to same unit

60. Ratios 1:2 and 2:4 are

a) not equal

b) equal

c) opposite

d) unrelated

Answer: b) equal

61. Equivalent ratios are obtained by

a) adding numbers

b) multiplying/dividing by same number

c) subtracting numbers

d) guessing values

Answer: b) multiplying/dividing by same number

62. Ratios 1:2, 2:4, 3:6 are

a) not equivalent

b) equivalent

c) doubled

d) halved

Answer: b) equivalent

63. Equivalent ratios have the same

a) denominator

b) value

c) numerator

d) name

Answer: b) value

64. To check ratios are equivalent, we write them as

a) fractions

b) decimals

c) whole numbers

d) words

Answer: a) fractions

65. To compare 1:4 and 4:8, first step is to

a) divide directly

b) write in fraction form

c) multiply directly

d) subtract

Answer: b) write in fraction form

66. If fractions are equal, then ratios are

a) equal

b) not equal

c) opposite

d) same

Answer: a) equal

67. Ratios 1:4 and 4:8 are

a) equivalent

b) not equivalent

c) similar

d) proportional

Answer: b) not equivalent

68. Comparing ratios uses

a) multiplication

b) fraction conversion

c) subtraction

d) none

Answer: b) fraction conversion

69. In comparing 1:3 and 1:4, who completed more work?

a) Reena

b) Ravi

c) Both equal

d) None

Answer: a) Reena

70. Ratio of 3m and 5m rope pieces is

a) 5:3

b) 3:5

c) 8:1

d) 1:8

Answer: b) 3:5

71. If $2:3 = 8:x$, then $x =$

a) 10

b) 12

c) 8

d) 9

Answer: b) 12

72. If $a:b = 3:5$ and $b:c = 5:7$, then $a:c =$

a) 3:7

b) 15:35

c) 3:5

d) 7:5

Answer: b) 15:35

73. If 4:6 and 10:15, they are

a) not equal

b) in proportion

c) opposite

d) cannot say

Answer: b) in proportion

74. If 6:9 and 8:12, then ratios are

a) equal

b) not equal

c) opposite

d) unrelated

Answer: a) equal

75. 7:2::21:6 means

a) equal ratios

b) not equal

c) opposite ratios

d) none

Answer: a) equal ratios

76. In $a:b::c:d$, $a/b =$

a) d/c

b) c/d

c) b/c

d) none

Answer: b) c/d

77. In $a:b::c:d$, $ad =$

a) bc

b) ab

c) cd

d) none

Answer: a) bc

78. If $a:b = 2:3$, then $b:a =$

a) 3:2

b) 2:3

c) 1:1

d) 1:2

Answer: a) 3:2

79. If $8x:13x = 8:13$, x is

a) any number except 0

b) 1 only

c) 0

d) 13

Answer: a) any number except 0

80. If $3x:6x$, ratio is

a) 1:2

b) 2:1

c) 3:6

d) 6:3

Answer: a) 1:2

81. The symbol '::' stands for

a) ratio

b) proportion

c) product

d) equality

Answer: b) proportion

82. When $ad = bc$, we can say ratios are

a) not proportional

b) proportional

c) opposite

d) half

Answer: b) proportional

83. The means in $4:6::8:12$ are

a) 4 and 12

b) 6 and 8

c) 4 and 8

d) 6 and 12

Answer: b) 6 and 8

84. The extremes in $4:6::8:12$ are

a) 6 and 8

b) 4 and 12

c) 4 and 8

d) 6 and 12

Answer: b) 4 and 12

85. In the ratio 1:2, multiplying both by 3 gives

a) 1:2

b) 3:6

c) 6:3

d) 2:1

Answer: b) 3:6

86. 9:12 reduced to simplest form is

a) 3:4

b) 4:3

c) 9:4

d) 12:9

Answer: a) 3:4

87. 25:100 simplified is

a) 1:4

b) 4:1

c) 1:25

d) 25:1

Answer: a) 1:4

88. If $a:b = 2:5$ and $b:c = 5:8$, find $a:c$.

a) 2:8

b) 10:25

c) 2:8

d) 2:8 simplified = 1:4

Answer: d) 1:4

89. A rope of 9m cut into 3m and 6m parts gives ratio

a) 6:3

b) 3:6

c) 1:2

d) 2:1

Answer: c) 1:2

90. If 4 pens cost ₹40, 1 pen costs

a) ₹4

b) ₹8

c) ₹10

d) ₹12

Answer: c) ₹10

91. If 8 apples cost ₹200, 2 apples cost

a) ₹25

b) ₹40

c) ₹50

d) ₹60

Answer: c) ₹50

92. The cost of 1 apple = ₹25, cost of 5 apples =

a) ₹100

b) ₹125

c) ₹150

d) ₹200

Answer: b) ₹125

93. If 3 bananas cost ₹30, 12 bananas cost

a) ₹60

b) ₹90

c) ₹120

d) ₹150

Answer: c) ₹120

94. Unitary method helps to

a) divide quantities

b) find one quantity first

c) find difference

d) none

Answer: b) find one quantity first

95. Ratio 2:3 written as fraction is

a) $\frac{2}{3}$

b) $\frac{3}{2}$

c) $\frac{1}{2}$

d) 2×3

Answer: a) $\frac{2}{3}$

96. If $1:3 = x:12$, then $x =$

a) 3

b) 4

c) 5

d) 6

Answer: d) 4

97. If $a:b = 2:3$, then $a = 10$, find b .

a) 12

b) 13

c) 15

d) 18

Answer: c) 15

98. If ratio of boys to girls is $4:5$, total 45 students, number of boys =

a) 15

b) 20

c) 25

d) 18

Answer: b) 20

99. If $5:8 = 10:x$, then $x =$

a) 12

b) 14

c) 16

d) 18

Answer: c) 16

100. If $a:b::b:c$, then

a) $a=c$

b) $a=b$

c) $b=c$

d) $a:b:c$ in continued proportion

Answer: d) $a:b:c$ in continued proportion